

ST LINA AGCM, AB, Canada

WMO: 715480

Lat: 54.2824N

Lon: 111.4512W

Elev: 619

StdP: 94.11

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.4	-28.5	-35.0	0.1	-31.2	-32.0	0.2	-28.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.2	17.2	25.6	16.6	24.0	15.9	19.1	24.7	17.9	23.4	16.9	22.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.0	22.4	15.7	12.0	20.7	14.7	11.2	19.4	57.0	24.9	53.1	23.3	49.7	22.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-35.3	30.2	2.8	1.2	-37.3	31.0	-38.9	31.7	-40.4	32.4	-42.5	33.3
			WB	-35.4	21.6	2.7	1.3	-37.4	22.5	-39.0	23.2	-40.5	24.0	-42.5	24.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-11.6	-12.3	-5.5	2.8	10.4	14.5	16.7	15.6	10.8	3.6	-5.5	-12.4	(6)
	DBStd	12.40	8.87	8.41	7.66	5.18	4.64	3.23	2.52	3.03	4.50	4.75	6.80	7.97	(7)
	HDD10.0	3461	668	624	479	221	53	6	0	2	43	204	464	696	(8)
	HDD18.3	5864	927	858	738	466	248	118	64	93	227	457	714	954	(9)
	CDD10.0	666	0	0	0	5	64	142	207	174	68	6	0	0	(10)
	CDD18.3	28	0	0	0	0	2	5	12	8	1	0	0	0	(11)
	CDH23.3	478	0	0	0	1	55	73	163	131	54	1	0	0	(12)
	CDH26.7	67	0	0	0	0	8	7	26	17	11	0	0	0	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	496	26	14	21	30	54	86	92	70	56	17	18	22	(15)
	PrecMax	708	68	61	36	54	127	186	163	167	143	61	45	40	(16)
	PrecMin	362	7	1	7	0	16	39	47	6	5	3	0	3	(17)
	PrecStd	88	15	12	10	15	30	43	36	40	35	14	12	11	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.9	11.8	22.1	27.8	27.6	29.3	28.7	28.4	21.5	11.9	5.6	(19)
		MCWB	3.0	3.0	5.5	10.3	13.8	16.2	19.4	18.3	16.4	12.5	6.2	1.7	(20)
	2%	DB	5.1	4.3	8.4	17.2	24.9	25.5	27.0	26.4	24.6	16.2	7.5	3.2	(21)
		MCWB	1.8	1.6	3.7	7.5	13.1	15.6	18.5	17.8	15.0	9.5	3.9	0.3	(22)
	5%	DB	3.2	1.9	5.9	14.1	22.1	23.6	25.1	24.7	22.0	13.3	5.1	0.7	(23)
		MCWB	0.8	-0.4	2.2	6.3	11.6	14.8	17.7	17.4	14.2	7.6	2.5	-1.4	(24)
	10%	DB	1.3	0.1	3.8	11.1	19.8	21.8	23.4	23.0	19.0	10.9	2.8	-1.6	(25)
		MCWB	-0.6	-1.4	0.8	4.8	10.4	14.1	16.6	16.2	12.4	6.4	0.9	-3.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.4	5.8	10.8	15.0	18.2	21.5	20.6	17.0	12.7	6.9	2.3	(27)
		MCDB	6.6	6.3	10.5	21.8	23.7	24.6	27.0	25.9	25.1	19.7	10.9	5.1	(28)
	2%	WB	2.1	1.6	3.8	8.1	13.7	16.9	19.6	19.0	15.6	10.1	4.4	0.3	(29)
		MCDB	4.6	3.9	7.8	16.0	22.4	23.0	25.0	24.7	23.3	14.5	7.1	3.0	(30)
	5%	WB	0.8	0.0	2.4	6.6	12.5	15.9	18.3	17.7	14.3	8.3	2.7	-1.3	(31)
		MCDB	3.1	1.7	5.6	13.1	20.4	21.7	23.7	23.1	21.3	12.3	4.9	0.6	(32)
	10%	WB	-0.7	-1.5	1.2	5.3	11.2	15.0	17.2	16.7	13.0	6.7	0.9	-3.1	(33)
		MCDB	1.3	0.2	3.4	10.4	18.2	20.1	22.3	22.0	18.1	10.5	2.6	-1.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.9	9.6	10.5	13.2	11.2	11.4	12.0	12.0	9.3	7.6	8.0	(35)
		MCDBR	9.4	10.8	11.0	15.2	16.5	14.5	14.6	15.1	17.1	13.9	9.9	9.5	(36)
	5% WB	MCWBR	7.4	8.7	7.2	8.0	7.4	6.6	7.5	7.6	8.5	8.2	6.7	7.7	(37)
		MCDWR	9.4	10.1	10.3	14.2	15.7	12.3	13.2	14.1	16.3	13.2	9.3	9.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.253	0.264	0.288	0.326	0.344	0.358	0.359	0.365	0.308	0.278	0.251	0.231	(40)
		taud	2.277	2.304	2.355	2.388	2.392	2.367	2.410	2.344	2.515	2.569	2.450	2.300	(41)
	Ebn at Noon		718	843	899	904	902	889	881	851	871	832	738	678	(42)
		Edh at Noon	56	77	93	104	110	114	107	108	80	60	47	44	(43)
(44) All-Sky Solar Radiation	RadAvg		0.88	1.92	3.45	4.66	5.70	5.84	5.95	4.84	3.40	1.90	0.89	0.62	(44)
	RadStd		0.09	0.15	0.25	0.39	0.47	0.41	0.32	0.38	0.36	0.30	0.11	0.06	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page